

Soil Health & Crop Recommendation Report

Report Date: November 24, 2025

Farmer Information	
Name:	raghu pujari
Phone:	8088117817
Location:	vijaypura
Land Size:	19.80 acres

Parameter	Value	Unit
Nitrogen (N)	80.0	mg/kg
Phosphorus (P)	50.0	mg/kg
Potassium (K)	50.0	mg/kg
pH	6.50	
Temperature	25.0	°C
Humidity	60.0	%
Rainfall	120.0	mm
Soil Type	Black Soil	

Soil Health Assessment

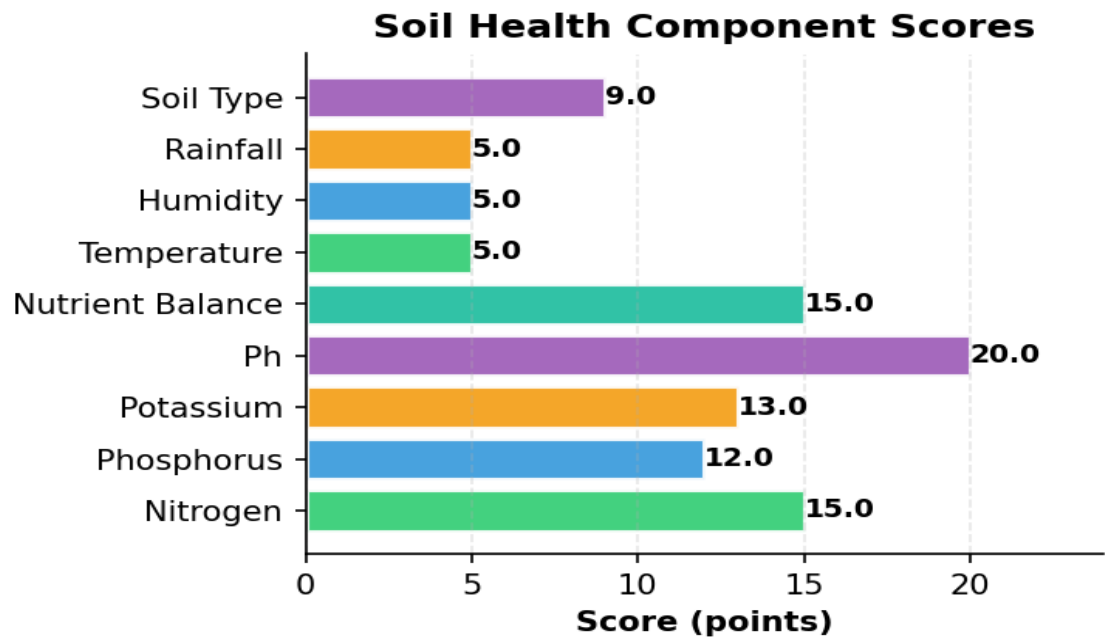
Overall Soil Health Score	99.0/100	Excellent	
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Component	Score	Points
Nitrogen	15.0	Points
Phosphorus	12.0	Points
Potassium	13.0	Points
Ph	20.0	Points
Nutrient Balance	15.0	Points
Temperature	5.0	Points
Humidity	5.0	Points
Rainfall	5.0	Points

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Soil Type	9.0	Points
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Weather Suitability for Cotton

Overall Suitability Score: 7.5/10 Good

- ✓ 25.0°C is ideal for cotton
- ✓ 60.0% is ideal for cotton — reduces disease risk
- 120 mm is below requirement (500-1000 mm); irrigation strongly recommended

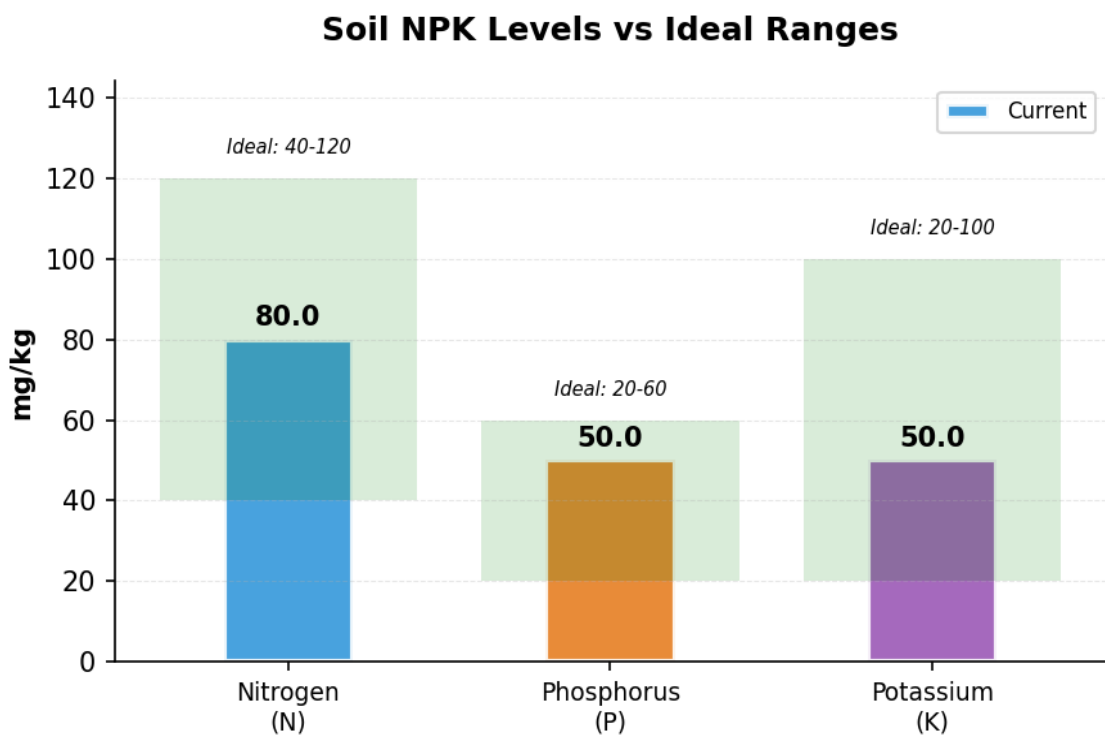
Recommended Crop

Predicted Crop	Cotton
Confidence Level	14.9%

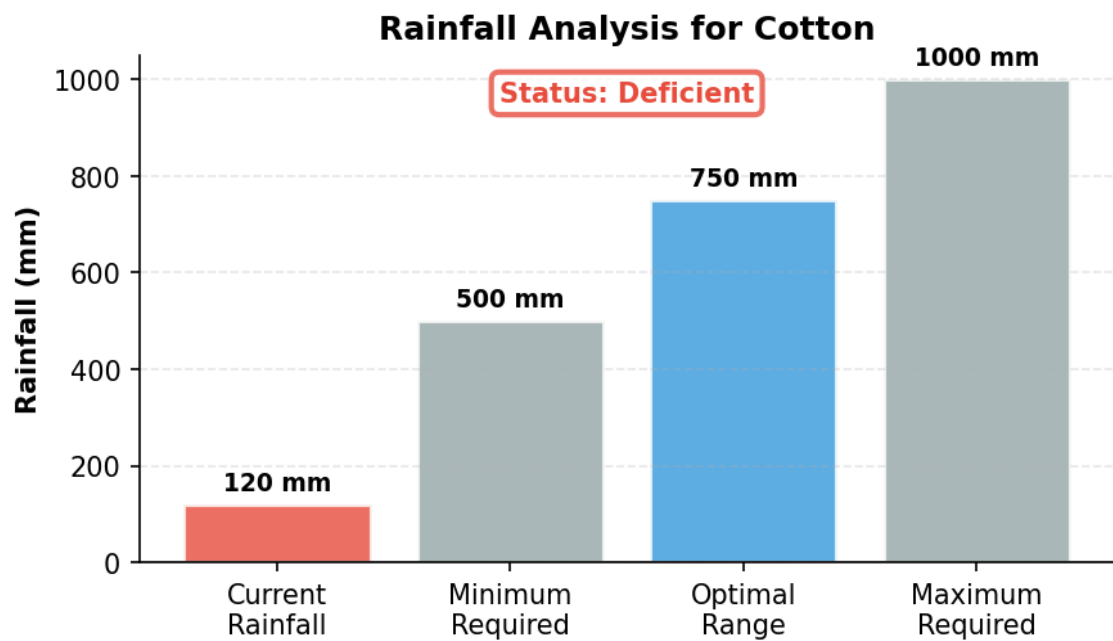
Soil NPK Levels Visualization

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Rainfall Analysis



Warnings & Observations

Action Required:

■■■■ below optimal → ■■■■■■■■ ■■■■■■■■

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Positive Indicators:

- ✓ ■■■■■■ level optimal
- ✓ ■■■■ level optimal
- ✓ ■■■■■■■■■■ level optimal
- ✓ pH ■■■■■■■■■■ (6.5)
- ✓ Soil type: ■■■■ ■■■■

■ Market Price Trends

Price Range	2000 - 3500 ■/Quintal
Average Price	2750 ■/Quintal

Best Market: Local Market, Your State (2750 ■/Quintal)

Market Advice: Market data integration is available in the system. In a production environment, this would show real-time market prices.

■ Soil Treatment Recommendations

- ■ Soil pH (6.5) is within optimal range (5.8-8.0) for cotton.
- ■ Nitrogen (80.0 mg/kg) is within optimal range for cotton.
- ■ Phosphorus (50.0 mg/kg) is within optimal range for cotton.
- ■ Potassium (50.0 mg/kg) is within optimal range for cotton.
- ■ High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 19.8 acres
Current Soil NPK: N=80.0 | P=50.0 | K=50.0 mg/kg
Optimal NPK for cotton: N=100 | P=45 | K=50 kg/acre

FERTILIZER APPLICATION SCHEDULE

■ **NITROGEN (N): 396.0 kg total (20.0 kg/acre)**

Application Method:

- Basal (50%): 198.0 kg at sowing
- First Top Dressing (25%): 99.0 kg at tillering stage
- Second Top Dressing (25%): 99.0 kg at flowering stage

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ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 99-198 tons

■ ■ Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

■ ■ Application: 59-99 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

■ ■ Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local agricultural extension recommendations for best results.

■ Crop-Specific Tips for Cotton

Avoid waterlogging; monitor for pests like bollworms. Requires warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

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ಮಣ್ಣಿನ ಆರೋಗ್ಯ ಮತ್ತು ಬೆಳೆ ಶಿಫಾರಸು ವರದಿ

ವರದಿ ದಿನಾಂಕ: November 24, 2025

ರೈತರ ಮಾಹಿತಿ	
ಹೆಸರು:	raghu pujari
ದೂರವಾಣಿ:	8088117817
ಸ್ಥಳ:	vijayapura
ಜಮೀನು ವಿಸ್ತೀರ್ಣ:	19.80 ಎಕರೆ

ಪರಮಿತರ	ಮೌಲ್ಯ	ಘಟಕ
ಸಾರಜನಕ (N)	80.0	mg/kg
ರಂಜಕ (P)	50.0	mg/kg
ಪೊಟ್ಯಾಸಿಯಂ (K)	50.0	mg/kg
pH	6.50	
ತಾಪಮಾನ	25.0	°C
ತೇವಾಂಶ	60.0	%
ಮಳೆ	120.0	mm
ಮಣ್ಣಿನ ಪರಕಾರ	ಕಪ್ಪೆ ಮಣ್ಣು	

ಮಣ್ಣಿನ ಆರೋಗ್ಯ ಮೌಲ್ಯಮಾಪನ

ಮಣ್ಣಿನ ಆರೋಗ್ಯ ಅಂಕ

99.0/100

ಅತಿಯುತತಮ

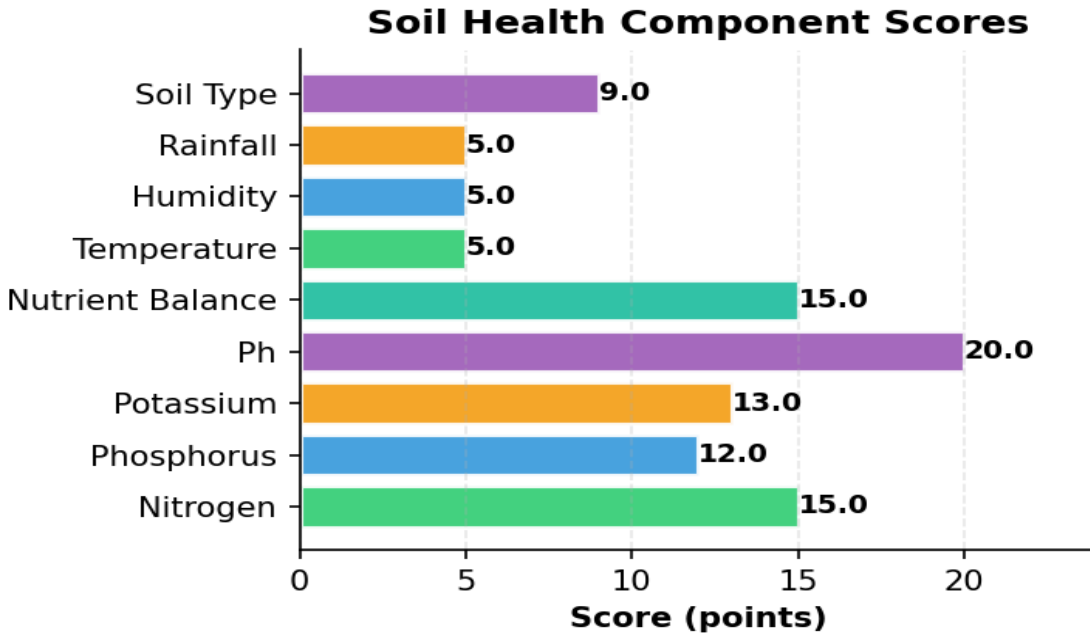


ಘಟಕ	ಅಂಕ	ಅಂಕಗಳು
Nitrogen	15.0	ಅಂಕಗಳು
Phosphorus	12.0	ಅಂಕಗಳು
Potassium	13.0	ಅಂಕಗಳು
Ph	20.0	ಅಂಕಗಳು
Nutrient Balance	15.0	ಅಂಕಗಳು
Temperature	5.0	ಅಂಕಗಳು
Humidity	5.0	ಅಂಕಗಳು
Rainfall	5.0	ಅಂಕಗಳು

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Soil Type	9.0	ಅಂಕಗಳು
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ಹವಾಮಾನ ಸೂಚಕತೆ Cotton

ಒಟ್ಟಾರೆ ಸೂಚಕತೆ ಅಂಕ: 7.5/10 ಉತ್ತಮ

- ತಾಪಮಾನ 25.0°C is ideal for cotton
- ತೇವಾಂಶ 60.0% is ideal for cotton — reduces disease risk
- ಮಳೆ 120 mm is below requirement (500-1000 mm); irrigation strongly recommended

ಶಿಫಾರಸು ಮಾಡಲಾದ ಬೆಳೆ

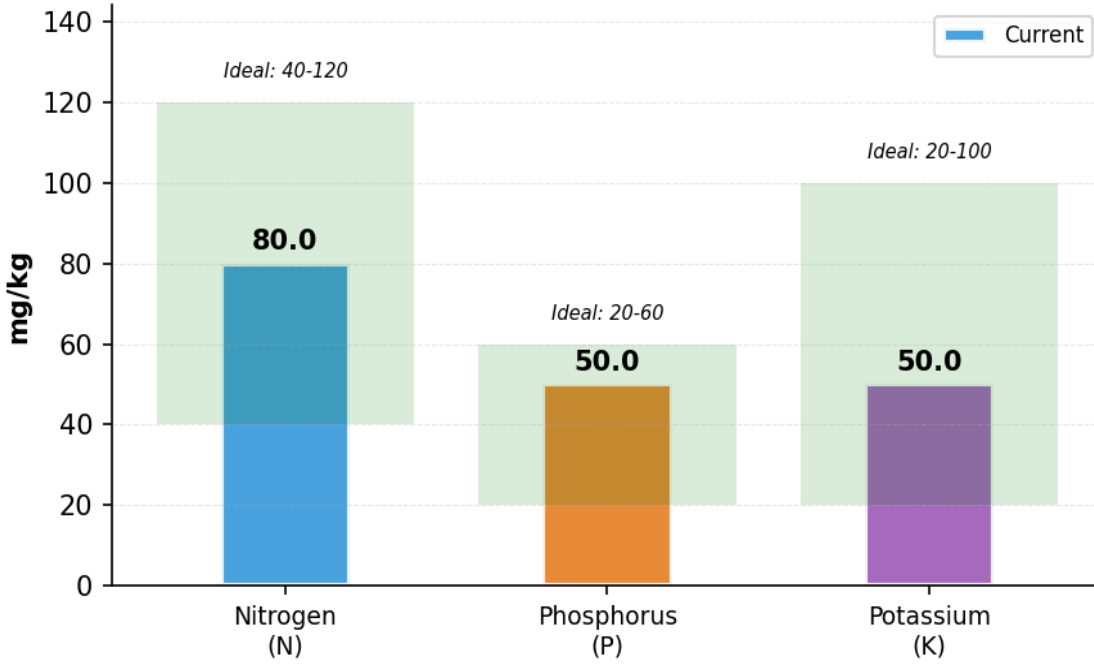
ಶಿಫಾರಸು ಬೆಳೆ	Cotton
ನಂಬಿಕೆ ಮಟ್ಟ	14.9%

Soil NPK Levels Visualization

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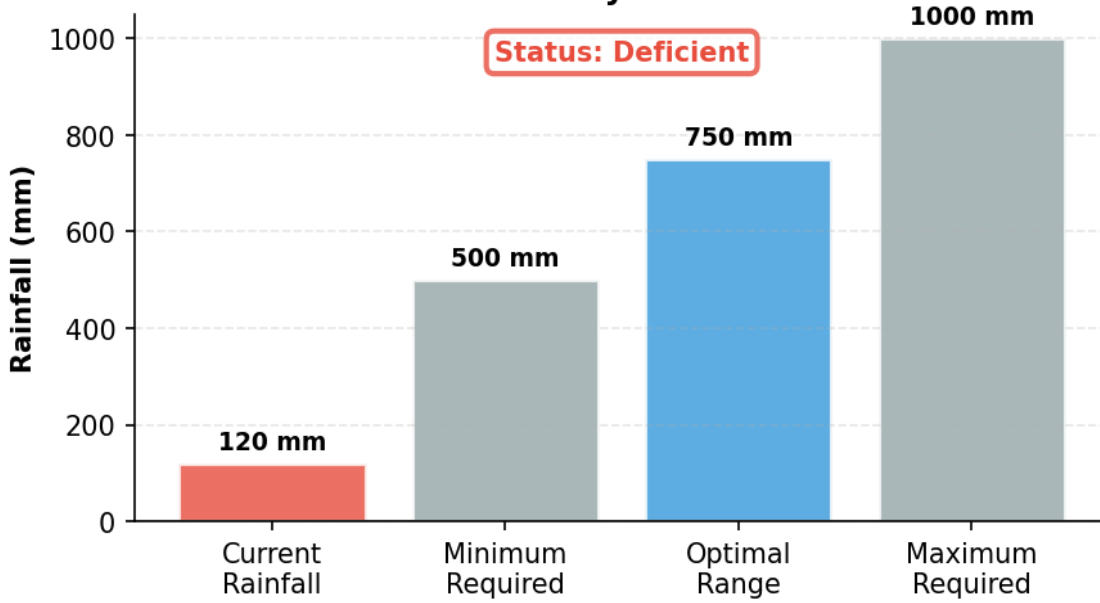
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Soil NPK Levels vs Ideal Ranges



ಮಳೆ ವೆಶಲೇಷನ್

Rainfall Analysis for Cotton



ಎಚ್ಚರಿಕೆಗಳು ಮತ್ತು ಅವಲೋಕನಗಳು

ಅಗತಯ ಕರಮ:

ಮಳೆ below optimal ☐ ನೇರವರೆ ಹೆಚ್ಚಿಸಿ

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ಸಕಾರಾತ್ಮಕ ಸೂಚಕಗಳು:

ಸಾರಜನಕ level optimal

ರಂಜಕ level optimal

ಪೊಟ್ಯಾಸಿಯಂ level optimal

pH ಉತ್ತಮವಾಗಿದೆ (6.5)

Soil type: ಕಪ್ಪು ಮಣ್ಣು

ಮಾರುಕಟ್ಟೆ ಬೆಲೆ ಪರಿಶೀಲನೆ

ಬೆಲೆ ವ್ಯಾಪ್ತಿ	2000 - 3500 ₹/Quintal
ಸರಾಸರಿ ಬೆಲೆ	2750 ₹/Quintal

ಉತ್ತಮ ಮಾರುಕಟ್ಟೆ: Local Market, Your State (2750 ₹/Quintal)

ಮಾರುಕಟ್ಟೆ ಸಲಹೆ: Market data integration is available in the system. In a production environment, this would show real-time market prices.

ಮಣ್ಣಿನ ಚಿಕಿತ್ಸಾ ಶಿಫಾರಸುಗಳು

- Soil pH (6.5) is within optimal range (5.8-8.0) for cotton.
- Nitrogen (80.0 mg/kg) is within optimal range for cotton.
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ಗೊಬ್ಬರ ಶಿಫಾರಸುಗಳು

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Avoid waterlogging; monitor for pests like bollworms. Requires warm climate.

ಹೆಚ್ಚುವರಿ ಟಿಪ್ಪಣಿಗಳು ಮತ್ತು ಹೆಚ್ಚು ನೆರವೇರಿಕೆ

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